Effects of Aggressive Versus Moderate Glycemic Control on Clinical Outcomes in Diabetic Coronary Artery Bypass Graft Patients

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Objective: This study sought to determine whether aggressive glycemic control (90-120 mg/dL) would result in more optimal clinical outcomes and less morbidity than moderate glycemic control (120-180 mg/dL) in diabetic patients undergoing coronary artery bypass graft (CABG) surgery.

Summary of Background Data: Maintaining serum glucose levels between 120 and 180 mg/dL with continuous insulin infusions decreases morbidity in diabetic patients undergoing CABG surgery. Studies in surgical patients requiring prolonged ventilation suggest that aggressive glycemic control (<120 mg/dL) may improve survival; however, its effect in diabetic CABG patients is unknown.

Methods: Eighty-two diabetic patients undergoing CABG were prospectively randomized to aggressive glycemic control (90-120 mg/dL) or moderate glycemic control (120-180 mg/dL) using continuous intravenous insulin solutions (100 units regular insulin in 100 mL: normal saline) beginning at the induction of anesthesia and continuing for 18 hours after CABG. Primary end points were the incidence of major adverse events (major adverse events = 30-day mortality, myocardial infarction, neurologic events, deep sternal infections, and atrial fibrillation), the level of serum glucose, and the incidence of hypoglycemic events.

Results: There were no differences in the incidence of major adverse events between the groups (17 moderate vs 15 aggressive; $P = 0.91$). Patients with aggressive control had a lower mean glucose at the end of 18 hours of insulin infusion (135 ± 12 mg/dL moderate vs 103 ± 17 mg/dL aggressive; $P < 0.0001$). Patients with aggressive control had a higher incidence of hypoglycemic events (4 vs 30; $P < 0.0001$).

Conclusions: In diabetic patients undergoing CABG surgery, aggressive glycemic control increases the incidence of hypoglycemic events and does not result in any significant improvement in clinical outcomes that can be achieved with moderate control. Clinical Trials.gov (ID #NCT00460499)

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Patients with diabetes mellitus requiring coronary artery bypass graft (CABG) surgery have increased perioperative morbidity and mortality, reduced long-term survival, and less freedom from recurrent ischemic events.[1–3] These results were thought to be irreversible because of impaired endothelial function, more diffuse coronary disease, and abnormal fibrinolytic and platelet function, which are more commonly seen in patients with diabetes mellitus.[4]

In a landmark study, Malmberg and coworkers[5] first demonstrated that maintaining serum glucose level less than 200 mg/dL in diabetic patients with an acute myocardial infarction (MI) using a continuous insulin infusion reduced mortality by 30%.[5] The results of this trial prompted our own group to perform a prospective randomized trial in 141 patients with diabetes mellitus undergoing CABG surgery to determine whether tighter perioperative glycemic control using continuous insulin infusions would improve perioperative outcomes.[6] The patients in whom glucose level was maintained between 120 and 180 mg/dL spent less time on the ventilator and required less inotropic support, had a significantly lower incidence of sternal wound infections and atrial fibrillation, and had a reduced length of hospital stay. Furthermore, these patients had significantly higher 5-year survival and less episodes of recurrent ischemia and achieved a lower angina class. Our findings were similar to those of Furnary and coworkers,[7–9] who demonstrated that maintaining serum glucose levels between 100 and 150 mg/dL using continuous insulin infusions decreased the incidence of deep sternal infections, lowered mortality, improved ventricular function, decreased the need for inotropic support, and shortened hospital stay. On the basis of these observations, the Society of Thoracic Surgeons Work Force on Evidence Based Surgery has recommended that continuous insulin infusions be used to keep serum glucose level less than 180 mg/dL in all patients during cardiac surgery and during their stay in the intensive care unit (ICU).[10]

Although there is general consensus that tighter glycemic control improves outcomes in diabetic CABG patients, the optimal target for serum glucose levels is unknown. Van den Berghe and coworkers,[11] in a study involving 1548 ventilated patients admitted to a surgical ICU, of which 62% had undergone cardiac surgery, found that mortality was significantly reduced in those patients requiring 3 days or more of ICU care if serum glucose level was maintained between 80 and 110 mg/dL.[11] However, recent trials in both ICU and non-ICU patients have raised concerns that more aggressive glycemic control may actually result in increased mortality from cardiovascular disease and increase the episodes of hypoglycemia.[12–20] This study was, therefore, undertaken to determine whether more aggressive glycemic control would result in more optimal clinical outcomes and less morbidity than can be achieved with moderate control in patients with diabetes mellitus undergoing CABG surgery.

METHODS

This study was approved by the Boston University Medical Center institutional review board (protocol #H-25760) and was registered with Clinical Trials.gov (ID #NCT00460499). An informed consent was obtained from each patient enrolled in the study.

Eligibility and Exclusion Criteria

This was a prospective, randomized single center trial involving patients with diabetes mellitus undergoing CABG surgery.

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Enrollment continued from September 2009 through February 2011. Patients with diabetes mellitus (controlled by insulin, oral medications, diet, or combination of oral medications and insulin) undergoing CABG surgery on cardiopulmonary bypass were included in the study. The patients were excluded if they had severe hyperglycemia (serum glucose > 400 mg/dL), which could not be controlled on a stable insulin regimen preoperatively if they had chronic renal failure (creatinine level ≥ 2.5 mg/dL), acute renal failure (urine output < 20 mL/hour for 3 hours), and those patients requiring concomitant procedures in addition to CABG surgery.

Study Protocol

Patients were randomly assigned to either a moderate group (serum glucose 120-180 mg/dL) or an aggressive group (90–120 mg/dL). After induction of general anesthesia, a continuous insulin infusion with 100 units of regular insulin in 100 mL of 0.9% normal saline was initiated at 3 mL/hour and titrated to maintain the targeted glucose level on the basis of the algorithm shown in Table 1. The study protocol continued during the periods of cardiopulmonary bypass and cardioplegic arrest, after the discontinuation of bypass, and for 18 hours in the ICU. After the 18-hour ICU period, patients were transitioned off the insulin drip using either short- or long-acting insulin agents and ultimately back to their preoperative diabetic regimens maintaining a fasting glucose level less than 120 mg/dL and 4 PM glucose levels less than 180 mg/dL.

Operative Techniques

Standard operative and anesthetic techniques were used on all patients. All procedures were performed using cardiopulmonary bypass with multidose infusions of antegrade, cold (4°C), blood (hematocrit = 20%), potassium (28 mEq/L), cardioplegia supplemented with mild systemic (35°C), and topical (cold saline lavage at 4°C) hypothermia. All vessels with at least 50% stenoses were bypassed and each patient received at least 1 internal mammary artery graft.

TABLE 1. Algorithm for Moderate and Aggressive Glycemic Control

Blood Glucose Result, mg/dL	Response
Moderate	
>240	Increase infusion by 2 units/hour and give 3 units IV regular insulin bolus
180–240	Increase infusion by 2 units/hour
121–180	No change
81–120	Decrease infusion by 2 units/hour
<80	Decrease infusion by 2 units/hour and give 25 mL 50% dextrose IV; recheck glucose in 30 minutes. If glucose ≥120, restart insulin infusion, but decrease by 1 unit/hour from previous infusion rate
Aggressive	
240–300	Give 6 units IV regular insulin bolus and increase infusion by 1 unit/hour
201–240	Give 3 units IV regular insulin bolus and increase infusion by 1 unit/hour
151–200	Give 2 units IV regular insulin bolus and increase infusion by 1 unit/hour
121–150	Increase infusion by 0.5 unit/hour
80–120	No change
60–79	Discontinue insulin infusion. Give 25 mL of D50% dextrose IV and recheck glucose in 30 minutes. If glucose >80 mg/dL, restart insulin infusion, but decrease by 1 unit/hour from previous infusion rate

In both moderate and aggressive groups, a continuous insulin infusion of 100 units of regular insulin in 100 mL of 0.9% normal saline was initiated at 3 mL/hour and titrated to the target range using the earlier algorithms.

Variables Measured

Measurements of heart rate, arterial blood pressure, pulmonary capillary wedge pressure, cardiac output, and cardiac index were monitored continuously from the time of anesthetic induction to 18 hours after arrival in the ICU. Serum glucose was measured every 30 minutes in the operating room and each hour in the ICU. Blood glucose measurements were obtained by means of arterial catheters and were measured with the use of point-of-care analyzers. Fatty acids were measured before induction and at 0, 6, and 18 hours after admission to the ICU. Primary end points of the study included the incidence of major adverse events (MAEs), changes in serum glucose, and the incidence of hypoglycemic episodes. Major adverse events consisted of 30-day mortality, the incidence of MI assessed by both enzyme and electrocardiographic changes as described previously,[6] the incidence of cerebral vascular accidents (CVAs), defined as a persistent neurologic deficit lasting 24 hours or more, the incidence of deep sternal wound infections defined as any infection reaching or directly involving the sternum,[21] and the incidence of atrial fibrillation that lasted for longer than 15 minutes. All patients received perioperative β-blockers and were continuously monitored on telemetry during their entire postoperative stay. A hypoglycemic event was defined as a blood glucose level less than 80 mg/dL. When this occurred, all patients had their infusions discontinued and were given 25 mL of D50 intravenous (IV) dextrose and had their serum glucose rechecked in 30 minutes. Infusions were restarted on the basis of their target glucose values as outlined in Table 1. Secondary outcomes included time on the ventilator defined as the time of admission to the ICU to the time of extubation, length of ICU, and hospital stay, all of which was standardized using "fast-track" protocols, weight gain defined as the difference between the weight the evening before surgery to 5 AM the day after surgery, an inotropic score that ranged from 0 to 4 (0 = no inotropic support; 1 = dopamine at 2 μg/kg for <24 hours; 2 = dopamine > 2 μg/kg for >24 hours; 3 = 2 inotropic agents; 4 = inotropic agents and intra-aortic balloon support), the amount of insulin administered, and the cardiac index during the study.

Statistical Analysis

Data were summarized using mean ± standard deviation for continuous variables and the number and percentage for categorical variables using statistical analysis systems (SAS, version 9.0). Statistical significance was defined as $P < 0.05$. χ^2 test and Fisher exact 2-sided tests were used to compare recurrences between the moderate and aggressive groups. Nonpaired Student t tests were used to compare measured data between the groups. Repeated-measures analysis of variance were used to measure changes in serum glucose, free fatty acids, and cardiac indices over different periods of time.

RESULTS

Eighty-two patients were eligible for randomization in this study and all 82 patients consented to participate. All patients completed the protocol and none were lost to follow-up. Patient profiles and risk factors are summarized in Table 2. The groups did not differ in age, sex, the urgency of surgery, type of diabetes control, hemoglobin A_{1c} (HbA_{1c}) levels, ejection fraction, or the incidence of hypertension, MI, congestive heart failure, or left main disease. All the patients received a β-blocker, statins, and aspirin before surgery and more than 50% in each group were taking angiotensin-converting enzyme inhibitors. The operative results are summarized in Table 3. Cross clamp and cardiopulmonary bypass times did not differ among the groups. The vast majority of patients in both groups had 3 to 4

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TABLE 2. Preoperative Patient Profiles

Variable	Moderate (n = 42)	Aggressive (n = 40)	*P*
Age, y	65 ± 9	63 ± 9	0.27
Sex (M/F)	26/16	32/8	0.09
Angina class IV, %	14 (33)	13 (33)	0.84
CHF, %	6 (14)	5 (13)	0.23
MI, %	24 (57)	23 (57)	0.65
HTN, %	42 (100)	37 (93)	0.59
Diabetes control, %			
Insulin	13 (31)	15 (38)	—
Oral	20 (48)	21 (52)	0.30
Diet	4 (10)	2 (5)	—
Insulin and oral	5 (11)	2 (5)	—
Hyperlipidemia	42 (100)	40 (100)	—
Surgical class, %			
Elective	9 (21)	13 (33)	—
Urgent	33 (79)	27 (67)	0.72
Emergent	0 (0)	0 (0)	0.23
HbA_{1c}	8.3 ± 1.5	8.4 ± 1.5	0.113
Ejection fraction	50 ± 9	49 ± 10	0.87
Left main disease	9 (21)	6 (15)	0.39
Cardioprotective medications			
β-Blockers, %	42 (100)	40 (100)	—
Statins, %	42 (100)	40 (100)	0.67
ACE inhibitors	25 (59)	24 (60)	—

Mean values are ± standard deviations. CHF indicates congestive heart failure; F, female; HTN, hypertension; M, male.

TABLE 3. Operative Results

Variable	Moderate (n = 42)	Aggressive (n = 40)	*P*
Cross clamp, min	55 ± 10	61 ± 15	0.12
Cardiopulmonary bypass, min	86 ± 14	95 ± 21	0.11
No. of Vessels bypassed			
2	10	6	0.30
3	26	23	—
4	6	11	—
LIMA, %	42 (100)	40 (100)	—

Values are mean ± standard deviation. LIMA indicates left internal mammary artery.

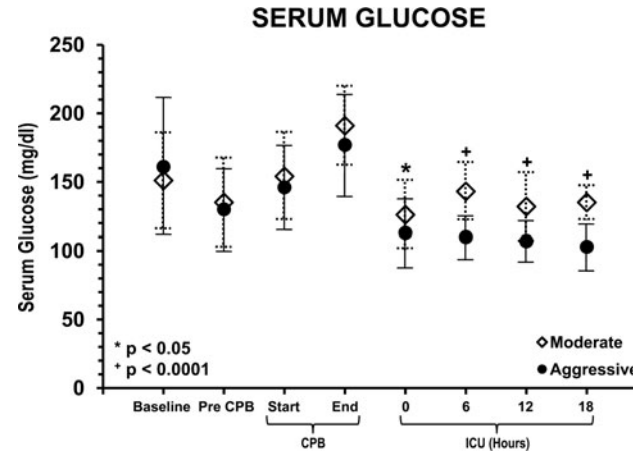

FIGURE 1. Serum glucose. A, Both groups had similar glucose levels at the time of anesthetic induction. B, Glucose levels were elevated in both groups during cardiopulmonary bypass but were maintained in the appropriate range for each group in the ICU. CPB indicates cardiopulmonary bypass.

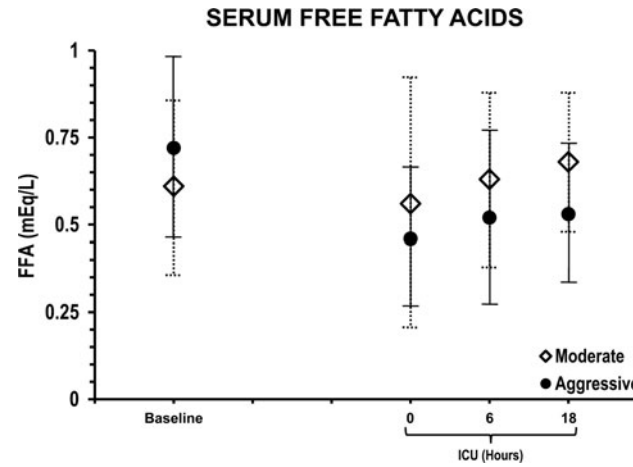

FIGURE 2. Serum-free fatty acids. A, Free fatty acids tended to increase during the period in the ICU in the moderate group but tended to remain lower in the aggressive group.

vessels bypassed. All the patients had a left internal mammary artery to bypass the left anterior descending artery.

The levels of serum glucose before, during, and after cardiopulmonary bypass for 18 hours in the ICU are shown in Figure 1.

Before anesthetic induction, there was no difference in serum glucose levels (151 ± 35 mg/dL, moderate vs 161 ± 50 mg/dL, aggressive; $P = 0.56$). Serum glucose increased in both groups during surgery. It was outside the targeted range at the end of cardiopulmonary bypass (191 ± 29 moderate vs 177 ± 37 aggressive; $P = 0.74$). However, both groups maintained target values during the 18 hours of infusion in the ICU. After 18 hours, serum glucose levels were significantly lower in the aggressive group (135 ± 12 mg/dL moderate vs 103 ± 17 mg/dL aggressive; $P < 0.0001$). Changes in free fatty acids are shown in Figure 2. Free fatty acids increased slightly in the moderate group from 0.61 ± 0.25 mEq/L before anesthetic induction to 0.68 ± 0.20 mEq/L after 18 hours in the ICU. In contrast, aggressive glycemic control resulted in a small but significant decrease in free fatty acids levels from baseline to the end of the infusion (0.72 ± 0.26 mEq/L to 0.53 ± 20 mEq/L; $P = 0.043$).

Major adverse events and hypoglycemic episodes are summarized in Table 4. There was no difference in the overall incidence of MAEs between the groups (17 moderate vs 15 aggressive; $P = 0.91$). The most common MAE was atrial fibrillation, which was similar between the groups (38% moderate vs 30% aggressive; $P = 0.59$). There were no 30-day mortalities or sternal infections among the groups. Three patients had a perioperative MI in the aggressive group: 2 by enzyme criteria and 1 had both enzyme and electrocardiogram changes. One patient in the moderate group had residual left-sided weakness after surgery. This resolved over 3 months. There was a difference, however, in the incidence of hypoglycemic events between the groups. Ninety percent of patients in the moderate group did not have a hypoglycemic episode as opposed to only 25% of aggressively treated patients ($P < 0.0001$). Seventy-five percent of

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TABLE 4. Major Adverse Events and Hypoglycemic Episodes

Variable	Moderate (n = 42)	Aggressive (n = 40)	*P*
Any MAE, %	17 (40)	15 (37)	0.91
30-day mortality, %	0 (0)	0 (0)	—
MI, %	0 (0)	3 (10)	0.23
CVA, %	1 (2)	0 (0)	0.47
Sternal infection, %	0	0	—
Atrial fibrillation, %	16 (38)	12 (30)	0.59
Hypoglycemic episodes, %			
0	38 (90)	10 (25)	<0.0001
1	3 (7)	16 (40)	—
>1	1 (3)	14 (35)	—

CVA indicates cerebrovascular accident; MI, myocardial infarction.

TABLE 5. Secondary Outcomes

Variable	Moderate (n = 42)	Aggressive (n = 40)	*P*
Time on ventilator, h	11.5 ± 9.5	12.0 ± 18	0.63
Weight gain, kg	10.6 ± 5.8	11.0 ± 7.5	0.71
Inotropic score >1, %	6 (14)	4 (10)	0.45
ICU, d	2.7 ± 0.5	2.9 ± 0.7	0.39
Hospital LOS, d	10.8 ± 3.5	10.1 ± 3.5	0.86
Amount of insulin, units	77 ± 30	118 ± 35	0.004

Values are mean ± standard deviation. LOS indicates length of stay.

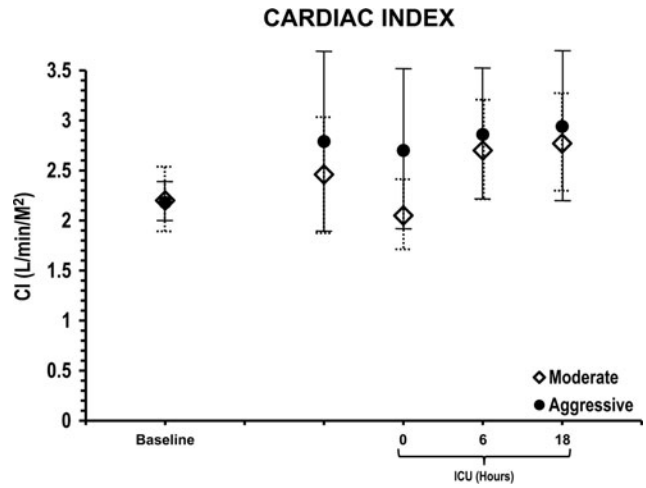

FIGURE 3. Cardiac index. A, Both groups maintained a cardiac index greater than 2 L/min/m^2 and there was no difference between the moderate and aggressive groups.

patients in the aggressive group had at least 1 hypoglycemic episode in contrast to 10% of moderately treated patients.

Secondary outcomes are summarized in Table 5 and Figure 3. Both groups had excellent postoperative hemodynamics as shown by the limited need for inotropic support and maintenance of cardiac indices more than 2.0 L/min/m^2 throughout the insulin infusion (Fig. 3). There was no difference between the groups in the time on the ventilator, weight gain, or length of stay in the ICU or hospital. The patients treated with aggressive glycemic control received significantly more insulin during the infusion (77 ± 30 units moderate vs 118 ± 35 units aggressive; $P = 0.004$).

DISCUSSION

Whereas studies have shown that maintaining serum glucose level less than 180 mg/dL is beneficial for improving outcomes in cardiac surgical patients, the potential for more aggressive control to reduce perioperative morbidity has been less clearly defined. In the study by Van den Berghe and coworkers,[11] cardiac surgical mortality was reduced only in those patients who received more than 3 days of ICU care and in those patients with multiorgan failure and sepsis. Aggressive glycemic control had no effect on morbidity or mortality in patients spending less than 3 days in the ICU. A subsequent study of aggressive insulin therapy in critically ill medical ICU patients by the same authors found no difference in hospital mortality between aggressive and moderate control.[22] However, patients who developed hypoglycemia had a significantly higher mortality rate. Two additional trials involving medical ICU patients failed to show any improvement in mortality with aggressive glycemic control and had to be discontinued early because of significant increases in hypoglycemic episodes.[14,23] In the normoglycemia in Intensive Care Evaluation-Survival Using Glucose Algorithm Regulation trial, 6104 patients expected to require 3 or more days of ICU care were randomized to an aggressive (81–108 mg/dL) or conventional (<180 mg/dL) glycemic group.[12] The 90-day mortality was significantly higher in the aggressive group (27.5% vs 24.9%; $P = 0.02$). Hypoglycemic episodes (<40 mg/dL) were more common in the aggressive group (6.8% vs 0.5%; $P < 0.001$) but no long-term complications were noted from these hypoglycemic episodes. The excess deaths in the aggressive group were predominantly from cardiovascular diseases.

Before 2009, most published studies in both outpatient and inpatient diabetic patients support aggressive glycemic control. Randomized, controlled trials in non-ICU hospitalized patients have conclusively demonstrated that the risk of long-term microvascular complications can be reduced by more aggressive glycemic control in patients with type 1 or 2 diabetes.[15,17,24] In the Diabetes Control and Complications Trial, there was a 60% reduction in the development or progression of diabetic retinopathy, nephropathy, and neuropathy between aggressive and moderate control.[25] However, the potential for aggressive control to reduce major cardiovascular events in these trials was less clearly defined.

These conflicting results prompted the initiation of several long-term trials to compare the effects of aggressive versus moderate glycemic control and cardiovascular outcomes in high-risk patients with diabetes mellitus. The Action to Control Cardiovascular Risk in Diabetes trial randomized 10,251 patients with either a history of a cardiovascular event or having significant cardiac risk factors to either aggressive or moderate glycemic control.[16] The aggressive group had significantly more weight gain and more episodes of severe hypoglycemia. In 2008, the trial was halted because of an increased rate of mortality in the aggressive arm (1.41% vs 1.14%) although the incidence of nonfatal MIs was reduced in the aggressive group. Patients with no history of cardiovascular events and those with a baseline HbA_{1c} less than 8% had a statistically significant reduction in the primary outcome (MI, stroke, and cardiovascular death). However, in both groups, patients who developed a hypoglycemic episode had a significantly higher mortality rate than those who did not.

In the Action in Diabetes and Vascular Disease: Preterax and Diamiciron Modified Release Controlled Evaluation trial, aggressive glycemic control significantly reduced the primary end point of microvascular (nephropathy and retinopathy) and major cardiovascular events (MI, CVA, and cardiovascular death) primarily due to a significant reduction in microvascular events.[18] There was no reduction in macrovascular events or overall cardiovascular mortality. In the

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Veterans Affairs Diabetes Trial, the patients receiving more aggressive glycemic control had no improvement in cardiovascular outcomes.[19] However, patients who developed hypoglycemia in both groups had a significantly higher incidence of developing adverse cardiovascular events, which included death, MI, CVA, or the need for coronary revascularization. However, post–ad hoc analyses suggested that patients who had diabetes less than 12 years might benefit from a more aggressive approach.

Why did these trials fail to show significant improvements in cardiovascular outcomes? Similar to our own patient cohort, many of these patients were already receiving cardiovascular preventive therapy with statins, angiotensin-converting enzyme inhibitors, aspirin, and weight-reduction programs. Hence, the added benefit of glucose lowering and reducing cardiovascular events may not have been as significant as anticipated. Furthermore, just as in our own earlier study, moderate glycemic control alone has beneficial effects on lowering serum glucose and HbA_{1c} levels that may be difficult to improve upon with more aggressive control. Finally, many of the patients in these trials had long-standing diabetes with a high atherosclerotic burden. Subset analysis of these trials suggested that patients with shorter durations of diabetes and lower HbA_{1c} levels might actually benefit from more aggressive control. Marik and Preiser[26] also hypothesized that aggressive management may be especially detrimental in patients not receiving parental nutrition. In their analysis, they concluded that the only variable that explained the reduced mortality seen in the Van den Berghe study was the increased use of intravenous nutritional support in patients who benefited from aggressive glycemic control. They hypothesized that the increased mortality seen in enterally fed patients receiving aggressive glycemic control is related to the adverse effects associated with hypoglycemia.

In all the earlier trials, hypoglycemia was more common in aggressively treated patients and was associated with an increased risk of all cardiovascular events whether or not patients were assigned to aggressive or moderate glycemic control.[20] Hypoglycemic episodes have been associated with sympathoadrenal activation and abnormal cardiac repolarization that may result in arrhythmias due to catechalomine surges.[27,28]) Hypoglycemia increases thrombogenesis, inflammation, and vasoconstriction, all of which result in an increased risk for mortality in patients with acute coronary syndromes.[29] Finally, hypoglycemia may act as a marker for advanced atherosclerotic disease in patients who are at an increased risk for cardiovascular events. It remains unclear as to whether hypoglycemia is a causative factor for increased cardiovascular events or simply reflects the presence of more severe disease.

On the basis of these observations, the American College of Physicians now recommends avoiding aggressive glycemic control in surgical intensive care unit /medical intensive care unit patients.[30] They recommend a moderate targeted blood glucose level of 140 to 200 mg/dL, which achieves glycemic control while avoiding hypoglycemia. Our own results are consistent with these recommendations and with a recent study by Bhamidipati and coworkers,[31] which showed that achieving moderate control (120-179 mg/dL) in isolated diabetic CABG patients was associated with the least amount of morbidity and mortality.[31] Although we have not shown any benefits to aggressive glycemic control, others have found that it may be beneficial during neonatal cardiac surgery by limiting the inflammatory response by decreasing cytokines and C-reactive protein and minimizing the reperfusion injury by reducing levels of troponin-1, fatty acid body proteins, and B-type natriuretic peptide.[32] Although the incidence of hypoglycemia was increased in the aggressive group, similar to our own data, there were no neurologic complications in these patients. Our study has also shown that aggressive control may be more effective in reducing perioperative inflammation by decreasing levels of free fatty acids. Long-term follow-up is planned to determine whether this reduction in inflammatory response translates into improved long-term outcomes.

Our study has several limitations. The results are derived from a single center and clinicians were not blinded to the treatment groups. This, however, would not affect the primary and secondary outcomes of the trial. Although the number of patients is small, the sample size was adequate to detect statistical differences in serum glucose, free fatty acids, hypoglycemic events, and the combination of MAEs. Despite the small sample size, our results are in keeping with the recent study by Bhamidipati and coworkers[31] in more than 1000 patients randomized to tight versus moderate control and the recent recommendations of the American College of Physicians on the basis of 14,768 patients. As the volume of CABG surgery continues to fall and the event rates continue to decrease because of better operative techniques, the use of cardioprotective medication before surgery, and improvements in perioperative care, the number needed to treat, rather than the actual sample size becomes more important in determining the clinical significance of the intervention. Hence, one would question the benefits of more aggressive glycemic control with its increased risk of hypoglycemia, if it required over a thousand patients to show a reduction in a single operative mortality.

We reported hypoglycemic events as serum glucose less than 80 mg/dL, which is a higher value than most trials. However, our patients remained intubated under anesthesia for more than 80% of the time that they received the insulin infusions and could not exhibit the usual signs and symptoms of hypoglycemia. In addition, given that point-of-care analyzers can differ from whole blood and plasma by 10% to 15%, less than 80 mg/dL threshold would capture all patients at risk. Hence, we elected to use a higher serum glucose to compensate for the inability to monitor the traditional signs of hypoglycemia.

Although our study has demonstrated that aggressive glycemic control provides no added improvement in clinical outcomes after CABG surgery in patients with diabetes mellitus, the concept of perioperative glycemic control remains important in reducing morbidity and enhancing long-term survival in these high-risk patients. Current data best support a glucose range of 120 to 180 mg/dL for patients undergoing CABG surgery. Although the optimal value for perioperative glucose control is still unknown, the concept of perioperative glycemic control using continuous insulin infusions is not debatable.

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DISCUSSANTS

D. Fullerton (Aurora, CO):

In your protocol, you deliberately chose to study patients for only the first 18 hours after the operation. Would a more protracted period of aggressive glucose control demonstrate some benefit?

Second, although it is true that the values of glucose between the 2 groups were statistically different, a mean of 103 in the aggressive group and 135 in the more moderately controlled group, and those values were clearly statistically different in the postoperative period, the absolute difference between 103 and 135 is relatively small. Do you think that a broader spectrum of differences might unveil some differences? For instance, what if the moderate group value had been closer to 170? Might that reveal an advantage of the more aggressive strategy?

Finally, it strikes me that perhaps the subgroup of diabetics who would most clearly benefit from an aggressive control strategy might be type I diabetics. In your 2 groups, about a third of the diabetics that you studied were type I diabetics. Have you studied whether focusing on just type I diabetics may reveal a benefit from a more aggressive strategy?

Response from H. Lazar:

In earlier studies, we found that extending the period of moderate control to more than 18 hours did not lead to any further improvement in clinical outcomes. Therefore, I do not feel that extending the period of aggressive glycemic control will result in any further improvement.

We found that even when we target a range of 120 to 180 mg/dL, most patients achieve values of 130 to 150 mg/dL. In our earlier studies, when we compared our moderate protocol of continuous insulin infusion with intermittent insulin boluses designed to keep blood glucose less than 250 mg/dL, we found that these patients averaged between 170 and 190 mg/dL and there was a difference in clinical outcomes. We learned from this that continuous insulin infusions result in better glycemic control than intermittent IV insulin boluses.

We have found that patients who are already receiving insulin do not benefit the most, because most of them come to us with well-controlled hemoglobin A_{1C} levels. We find that the patients who benefit the most are those who have just recently been diagnosed with diabetes and those with elevated glucose values during surgery who have not been diagnosed with diabetes but were found to have elevated HbA_{1c} levels preoperatively. Indeed, historical studies have shown that these are the patients at highest risk because they have never been treated before and carry a high atherosclerotic burden. This is the group that benefits most from this therapy.

DISCUSSANTS

R. Udelsman (New Haven, CT):

In the work presented today, you showed over a 4 times higher probability of hypoglycemia in your aggressively treated group, although you did not see associated morbidity. Going forward, continuous glucose monitoring devices will likely be used in our operating rooms and ICUs. Have you considered using continuous glucose monitoring devices in your population of patients?

Response from H. Lazar:

As part of this study, we did try a continuous glucose monitor. We, and others, have found that they are not as accurate because insulin does not take effect right away when treating a glucose value. It is not like using sodium nitroprusside to control blood pressure.

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Continuous glucose monitoring has not been helpful, especially in the hour-to-hour regulation of glucose that is required in the ICU.

This study and all other studies involving cardiac surgery patients have not shown that hypoglycemic events result in any increased morbidity and certainly not any neurologic events. In other studies, where hypoglycemia has been a problem, it has resulted in increased cardiovascular events in patients who are not undergoing CABG surgery. My own theory for that is that hyperglycemia is a surrogate marker for patients with increased atherosclerotic burden, and patients who develop hypoglycemia in a nonsurgical setting on these trials or in the medical intensive care unit probably have underlying coronary disease. Hypoglycemia results in changes in sympathetic tone that affects changes in vasoconstriction and vasodilatation in patients who have underlying coronary disease and have not been revascularized. Those patients get in trouble because they have a potentially ischemic myocardium.

DISCUSSANTS

G. Bochicchio (Baltimore, MD):

When I designed trials of this same sort, the number of patients has been between 600 and 700, 300 in each arm. Please comment on your power analysis.

Second, how did you measure glucose? There is a difference between meters that are primarily used in the operating rooms versus meters that are used in the ICU. Several studies have demonstrated that the error rate when using a meter may be as high as 20%, depending on your meter and your hematocrit. How did that factor into your research?

Finally, you reported on mean glucose ranges, but in the glucose world, what is reported is the percent of values in the box; that is, if you are going to go to 90 to 120, whether it is 70% of the time, 90% of the time, and so on. When I look at your numbers, I do not see a difference in the reported glucose values between moderate and tight, and with all the variables in meters, the Roche meter versus the AccuCheck versus the ABL, standard deviations can be 20%, which almost equalizes the differences.

Response from Harold Lazar:

First, regarding the power analysis, the end points were chosen to account for changes in MAEs. And they were cumulative because, fortunately, in patients undergoing coronary artery bypass surgery today, adverse outcomes have decreased. That was 1 surrogate and another was to use the end point of serum glucose to see what the absolute differences were. The studies were adequately powered for that and for the incidences of hypoglycemic events. This study is based on 82 patients. And in the cardiac surgery community, as volume decreases, the events become even fewer; you would need thousands of patients to see a difference in mortality. Even though you may reach statistical significance as the number needed to treat grows larger, this may not be clinically significant. Our initial studies showing the benefits of moderate glycemic control were based on 141 patients and were substantiated by other, larger studies. It is like using a poll to predict an election. Although 30 million to 40 million people vote for the president of the United States, a well-conducted poll that looks at only a fraction of those voters, examining the right parameters, can predict the eventual winner.

As far as how we measured glucose, there are a number of methods. We used point-of-care testing and as you point out, there can be as much as a 20% variance. But testing was standardized for all the patients using arterial lines to obtain the samples. That is one of the reasons why we defined a hypoglycemic episode as a glucose value of less than 80 mg/dL to account for those changes.

Finally, although in the ICU, better than 90% of patients in both groups reached their targeted goal, which was 90 to 120 in the aggressive group and 120 to 180 in the other group. The only exception was during surgery, when the goal was achieved significantly more often in the moderate group than it was in the aggressive group. That demonstrates the difficulty of maintaining glycemic control in the operating room, when you are accounting for insulin resistance on bypass, use of glucose-containing solutions, and, in some instances, use of inotropes.

DISCUSSANTS

A. Galloway (New York, NY):

What I find appealing is the potential for tighter glucose control to impact lymphocyte killing power and endothelial cell function, which could potentially affect late graft patency. Do you have any data on these tighter end points of lymphocyte killing power or endothelial cell function? Do you plan to do any other studies along these lines?

Response from H. Lazar:

Specifically looking at endothelial cell functions?

DISCUSSANTS

A. Galloway (New York, NY):

Yes, the potential benefit for tighter glucose control would likely be to prevent sepsis using a surrogate end point of white cell killing power or of endothelial cell function that might plausibly affect late graft patency. Have you done any studies on white cell killing power or endothelial function comparing the 2 groups?

Response from H. Lazar:

Before carrying out the clinical aspects of this work, we worked a great deal with glucose and insulin in the pig model of acute coronary occlusion and reperfusion. We showed that complement and endothelial function, both in the arteries at risk and in the other remote areas of the myocardium, were markedly improved in animals that received insulin. One of our theories is that improved endothelial function is seen with the use of insulin during this period of time. We hypothesized that the reason why, long term, the group of patients fared better because of enhanced graft survival.

Studies done in patients undergoing acute percutaneous coronary interventions that looked at some inflammatory markers during the time of insertion of stents found that those patients who achieved glycemic control not only experienced a lower inflammatory response but, ultimately, also had better stents patency. In the clinical trials, we looked at free fatty acids, both in our earlier studies and in this current study, and it is important for us to follow up on these patients 3 to 5 years down the line to see whether or not the aggressive group saw any improvement as far as recurrent symptoms of ischemia or the need for re–revascularization are concerned.

DISCUSSANTS

A. Galloway (New York, NY):

Is there evidence that a glucose level of 90 is better than 140 in terms of endothelial cell function, or is that difference from tight versus moderate control not supported by preliminary animal data?

Response from H. Lazar:

I am not aware of any animal data that have looked at this issue, but the van den Berghe trial found that inflammatory markers were markedly decreased with tighter glucose control using their more aggressive regimen. However, this did not necessarily translate into improved clinical outcomes.

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